1. Document History

Date	Version	Description	Author
2023-04-03	0.1	first draft	Edmund Widl (AIT)
2024-01-25	0.2	Added identification	Filip Pröstl Andrén (AIT)

2. Control Function Identification

Control Function Identification	
Control Function ID	Name
`MENB-CF-PID`	PID controller for the hydraulic pump

3. Functional Description

Functional Description
Background: System configuration `MENB-SC` includes a heat pump. A hydraulic pump controls the mass flow in the heat pump's condenser loop, where cold water is drawn from the bottom of a storage tank, heated up in the heat pump's condenser and fed back at the top of the tank. **Problem formulation:** The operation of the heat pump is limited by a power consumption setpoint (coming from a voltage controller). The controller specified in this document regulates the mass flow rate through the heat pump's condenser loop via the hydraulic pump such that this power consumption setpoint is met.

4. Terminology (optional)

Terminology	
PID controller	A proportional-integral-derivative (PID) controller is a control loop mechanism employing feedback that is widely used in applications requiring continuously modulated control. A PID controller continuously calculates an error value $e(t)$ as the difference between a desired setpoint (SP) and a measured process variable (PV) and applies a correction based on proportional, integral, and derivative terms (denoted P, I, and D respectively).

5. Methodology

Methodology
This control function implements a PID controller.

6. Limitations

Limitations
N/A

7. Inputs

Control Function Input	
Name	meas_hp_p_el
Type (according to SC description)	MeasureConnection
Unit	kW_{el}
Range	[0, 100]
Expected update rate	1 Hz
Description	measurement of electrical power consumption of heat pump

Control Function Input	
Name	setpoint_hp_p_el
Type (according to SC description)	CtrlConnection
Unit	MW_{el}
Range	[0, 0.15]
Expected update rate	every 5 min
Description	setpoint for power consumption of heat pump from voltage controller

8. Outputs

Control Function Output	
Name	setpoint_cond_mflow
Type (according to SC description)	CtrlConnection
Unit	kg/s
Range	[0, 10]
Expected update rate	1 Hz
Description	setpoint for heat pump condenser mass flow

9. Use Cases

Use Case Example	Heat pump operation
Date created	2023-04-03
Actor	• PID controller • hydraulic pump • heat pump

Description	The mass flow through the heat pump's condenser loop is determined by the operation of the hydraulic pump. With the help of a PID controller (see Figure 1) the mass flow of the hydraulic pump is regulated.
Preconditions	The power consumption setpoint for the heat pump is not zero.
Postconditions	N/A
Priority	medium
Frequency of use	1 Hz
Normal course	The mass flow through the hydraulic pump is regulated using the heat pump's power consumption as process variable and a setpoint (see Figure 1). PID controller parameters: • K_p = 1e-4 • T_i = 10 s • T_d = 10 s
Alternative course	In case the power consumption setpoint of the heat pump is set to zero, the hydraulic pump is switched off. The PID output is then set to zero, resulting in zero mass flow rate.
Exceptions	N/A
Assumptions	N/A
Notes and issues	N/A

10. Diagrams

Diagrams
Schematic view of the control setup

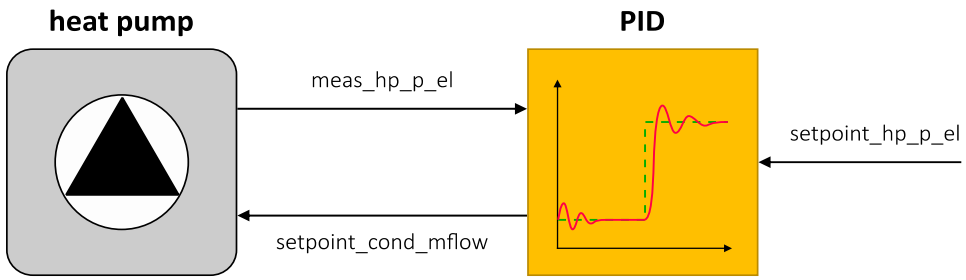


Figure 1: PID control setup for regulating the heat pump's power consumption.

11. Algorithmic Functions (pseudocode)

Algorithms
Controller algorithm
N/A

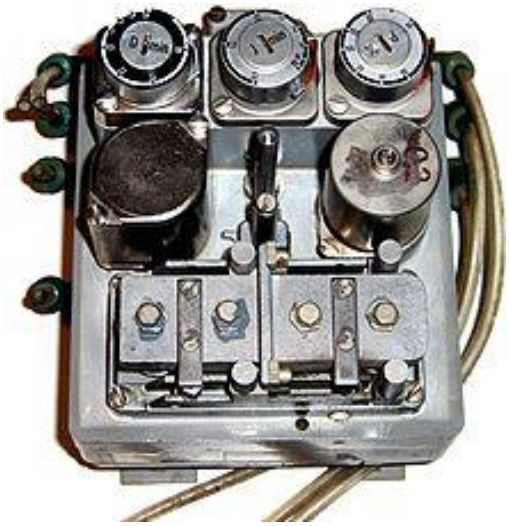
12. Deterministic Functions

Deterministic Functions
Error value: $$e(t) = \texttt{meas_hp_p_el}(t) - \texttt{setpoint_hp_p_el}(t)$$ **Control variable**: $$\texttt{setpoint_cond_mflow}\,(t) = K_\mathrm{p}\left[e(t) + 1/T_\mathrm{i}\int_0^t e(\tau)\mathrm{d}\tau \quad + T_\mathrm{d}\frac{\mathrm{d}e(t)}{\mathrm{d}t}\right]$$

13. Stochastic Functions

Stochastic Functions
N/A

14. Deployment

Deployment
A state-of-the-art pneumatic PID controller device is used for the deployment (see Figure 2).  Figure 2: PID controller device (source: Wikipedia)

15. References

References
N/A